

Epistemic Decoupling in Human and AI Proofs

Same Theorem, Different Interior

Working paper

1 August 2026

Abstract

Two correct proofs of the same theorem can be built very differently. We ask whether those architectural differences also change the epistemic support that a proof network provides. The question extends Viteri and DeDeo’s account of epistemic phase transitions, in which converging deductive and abductive pathways allow confidence to survive local fallibility.

We compare 2,321 theorem-matched human- and AI-provenance Lean scripts and, in a separately filtered sample, 312 matched elaborated proof-term networks. The theorem strongly constrains both populations: human and AI proof sizes correlate at $\rho_s = 0.67$ in source and $\rho_s = 0.76$ after elaboration. Within that common frame, AI proofs have a larger interior. They contain 1.41 times as many local claims per tactic and 1.12 times as many non-constant term nodes per distinct library constant. Their named interface changes much less. Yet the Viteri–DeDeo model assigns nearly the same theorem belief to both populations, and interior expansion has little association with belief change ($\rho_s = 0.046$).

We call this pattern *epistemic decoupling*: proof architecture can change without a commensurate change in modeled epistemic outcome. The result is not a claim that all proofs are equally understandable. It identifies the missing scientific object: the interface through which a checked certificate becomes available to a reader and a mathematical community. For AI-for-math, the next problem is therefore not only proof search, but proof design.

1 Introduction

Two proofs can agree on the theorem and disagree on nearly everything else. They can choose different lemmas, introduce different intermediate claims, and distribute their work across different levels of abstraction. In informal mathematics these differences are familiar. In a proof assistant they become measurable, because both constructions can be checked against exactly the same formal statement.

This creates a new problem for the study of mathematical reasoning. When correctness is held fixed, which features of a proof remain epistemically important? The question is practical as well as theoretical. AI systems can now produce large numbers of kernel-checked proofs (Yang et al., 2023; Ren et al., 2025; Hubert et al., 2026). Correctness is indispensable, but abundance makes selection unavoidable. We must decide which certificates are worth reading, maintaining, and reusing.

We propose a simple answer and give it a name. *Epistemic decoupling* occurs when alternative proofs of the same theorem differ substantially in architecture but not in an epistemic outcome. In the present study, architecture is the amount of internal derivational structure placed behind a proof’s named contact with the library. The epistemic outcome is theorem belief under the network model of Viteri and DeDeo (2022). The term is deliberately narrow. It describes a relation between a structural contrast and a specified epistemic functional; it does not declare proof structure irrelevant in general.

The paired human–AI comparison reveals three parts of the pattern. First, theorem identity imposes a common frame: human and AI proof sizes are strongly associated. Second, AI provenance tracks a change inside that frame: AI proofs contain more local and elaborated structure per named interface element. Third, this interior expansion neither produces a detectable reduction in modeled theorem belief nor clearly predicts the small human–AI belief difference. Architecture moves; the modeled endpoint does not move with it.

This result advances the phase-transition account in a controlled setting. Viteri and DeDeo showed how a network of mutually supporting claims can sustain confidence despite fallible local inference. Their historical and Coq examples differed in theorem, scale, representation, and provenance. Here the theorem is fixed within every comparison. The question is no longer only whether one proof network can sustain belief, but whether that belief is robust to another realization of the same formal target.

The paper proceeds in that order. We first define epistemic decoupling and the interface–interior distinction. We then observe one paired comparison at two resolutions: the source script and the elaborated proof term. The results follow a single sequence—common constraint, interior change, epistemic invariance. We then consider what the pattern means for cognition, philosophy of proof, and AI proof design. [Section A](#) returns to the original Coq analysis and asks directly what the human Lean evidence confirms.

2 Epistemic decoupling

2.1 From one proof to a paired contrast

Let S be a formal statement and let P_H and P_A be checked proof terms of S with human and AI provenance. From a term P we extract a directed acyclic expression graph $G(P)$. Constant nodes refer to named declarations in the ambient library. We call this named frontier the proof’s *interface*. Applications, abstractions, variables, types, and other non-constant expressions form its *interior*.

The distinction is operational, not metaphysical. A library constant is not necessarily a deep mathematical idea, and a non-constant node is not necessarily a trivial implementation detail. The boundary asks a narrower question: how much internal expression structure is placed behind each named point of contact with the surrounding library? We measure that load by

$$\chi(P) = \frac{|V(G(P))| - |C_V(P)|}{|C_D(P)| + 1}, \quad (1)$$

where $C_V(P)$ is the set of constant vertices in the structurally deduplicated graph and $C_D(P)$ is the set of distinct library declarations they name. At source level, explicit local `have` claims per tactic provide an independent, visible proxy for the same idea.

Let $B_\epsilon(P)$ be theorem-node belief under a network model with local error rate ϵ . For each theorem pair, define

$$A = \log \frac{\chi(P_A)}{\chi(P_H)}, \quad E = B_\epsilon(P_A) - B_\epsilon(P_H). \quad (2)$$

The empirical signature of decoupling is a systematic architectural contrast A together with a small epistemic contrast E and little association between the two. This definition makes the claim falsifiable. It would fail if larger AI interiors reliably produced lower theorem belief, or if the structural contrast vanished under elaboration.

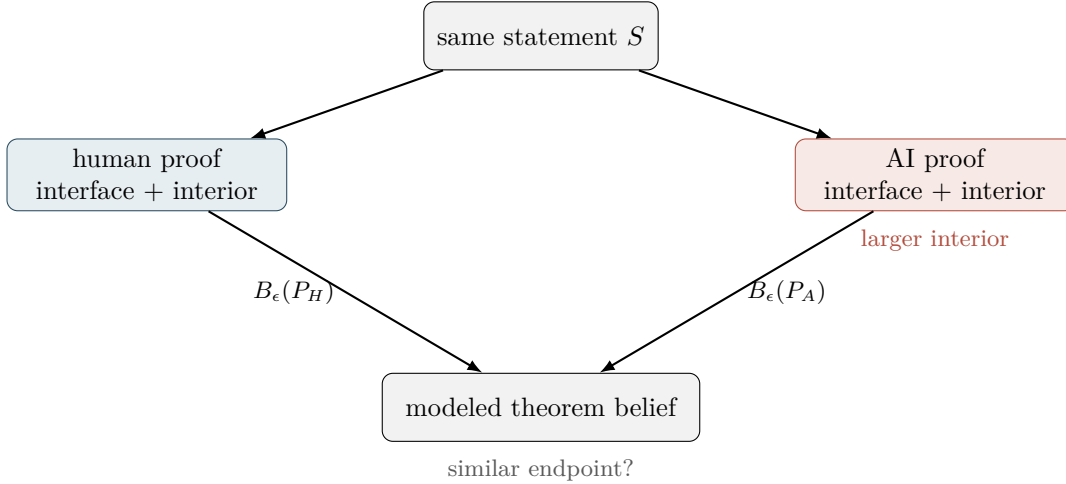


Figure 1: The design of the argument. Pairing fixes the theorem. The analysis then asks whether a change in proof interior is accompanied by a change in modeled theorem belief.

2.2 Why a network can decouple size from belief

A linear chain gives one intuition about proof reliability. If every link can fail independently, each additional step is another liability. A network gives another. Claims can be reused; independent routes can converge; and a conclusion can provide abductive support for premises that also support it. Under the Viteri–DeDeo dynamics, path proliferation can offset the loss of influence along any single path. A sufficiently connected proof can therefore cross from local uncertainty to high global confidence (Viteri and DeDeo, 2022).

This mechanism makes decoupling possible but does not guarantee it. Adding nodes could create redundant support, extend fragile chains, or merely expose compiler bookkeeping. The paired design asks which of these possibilities dominates for current human- and AI-provenance Lean proofs. It also exposes a deeper issue: kernel expressions need not be the units on which a human reader reasons. If term-level expansion leaves network belief nearly unchanged, the cognitively consequential variation may lie in how those expressions are grouped and presented.

3 One comparison at two resolutions

3.1 The paired corpus

The source data come from NuminaMath-LEAN proof artifacts (AI-MO, 2025). Each retained record supplies one formal statement, one human-provenance proof, one AI-provenance proof, validation metadata, and a coarse source label. Both proofs must be marked valid and available, contain a measurable proof body, and expose at least one premise-like reference under the corrected lexical parser. The source sample contains 2,321 theorem pairs from 12 source groups.

“Human” and “AI” refer to dataset provenance, not pure causal treatments. Human editing, retrieval, shared training data, and selection can blur the categories. The value of the design is more basic: within each row, statement, hypotheses, and ambient formal environment are shared. The comparison therefore removes theorem identity, a major confound in cross-corpus studies.

We observe the same conceptual contrast at two resolutions. Table 1 gives the map. The source and term samples are filtered separately, so evidence should be combined conceptually, not by

Table 1: One comparison at two resolutions. The source layer records visible proof organization; the term layer records the elaborated certificate.

Resolution	Paired sample	Interior measure	Interface measure
Source	2,321 statements	Explicit local have claims per tactic	Distinct premise-like identifiers after removing comments and strings
Term	312 statements	Non-constant expression nodes per distinct constant	Distinct named library constants

pretending that every estimate comes from one common table.

3.2 Visible organization in source

For each script we count lines, tactic invocations, and explicit **have** declarations. We also use two deliberately conservative lexical measures of named premise contact: a loose count of distinct dotted or capitalized identifiers and a strict count of dotted identifiers. Comments and strings are removed first. These counts are not dependency graphs. Their role is to check whether an apparent interface difference is visible before elaboration and whether it survives a change in token rule.

To retain proofs with no explicit local claims, the source interior proxy is

$$\chi_R = \frac{n_{\text{have}} + 1/2}{n_{\text{tactic}} + 1}. \quad (3)$$

Because this ratio uses pseudocounts, raw medians and paired differences are reported alongside it.

3.3 Certificate structure after elaboration

Lean elaboration turns each script into a kernel-checkable term. We deduplicate structurally equal expressions to form a DAG and direct edges from child expressions toward the expressions that contain them. The graph includes the theorem type and proof value. Named constants remain at the frontier; imported declarations are not recursively expanded. This **term0** representation is therefore the root certificate behind a fixed interface, not all of Mathlib.

Both sides were extracted for 494 eligible pairs. We exclude a pair if either graph contains **sorryAx** or **SyntheticOpaque**, which mark Lean error recovery rather than a completed proof. The clean term sample contains 312 pairs. This 36.8% pairwise exclusion is substantial, and the retained terms should not be treated as a random sample of all valid source proofs. The source and term samples overlap on 184 statements.

3.4 The epistemic endpoint

For each clean term graph we implement the symmetric-error case of the asymmetric-Ising model in [Viteri and DeDeo \(2022\)](#). Nodes begin true with prior probability 0.75. Deductive and abductive couplings correspond to local error rates $\epsilon \in \{.10, .05, .01\}$. We run ten sweeps and 32 Monte Carlo chains per graph, using common seeds within each theorem pair. Every retained graph is one weak component containing the theorem root, so disconnected isolates do not affect this comparison.

The primary endpoint is theorem-node belief averaged across the three error settings. It is a model implication, not a measurement of a reader. This distinction matters throughout: the study can establish decoupling inside the network model and use that result to motivate, but not replace, a cognitive experiment.

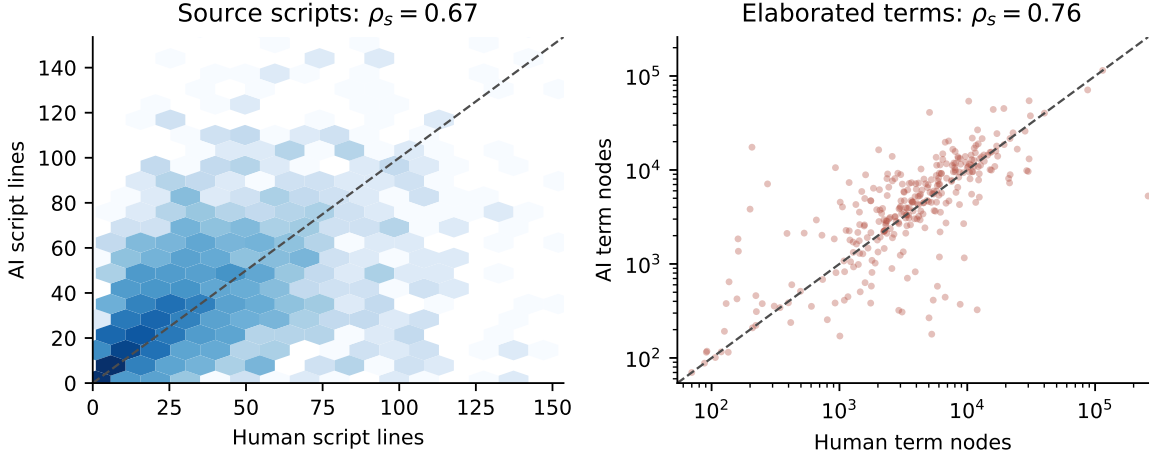


Figure 2: The theorem supplies a common architectural frame. Each point is a same-statement human/AI pair; the dashed line is parity. The source panel uses density shading because the sample is large. Coefficients are paired Spearman correlations.

3.5 Paired uncertainty

We report medians of paired differences or paired log ratios. Spearman correlation measures how strongly theorem identity constrains proof scale and how the two paired contrasts co-vary. The corpus contains many related competition problems, so ordinary proof-level intervals would be too narrow. Our intervals resample the 12 coarse problem-source groups with replacement, retaining all observations in each selected group. The analysis uses 20,000 resamples with a fixed seed. An identical ORCHARD run reproduced the local numerical summary byte for byte.

4 The decoupling pattern

4.1 First, the theorem supplies a common frame

Human and AI proofs of the same statement are not independent constructions. Their script line counts correlate at $\rho_s = 0.67$ (95% source-cluster interval $[0.59, 0.74]$), and their elaborated node counts correlate at $\rho_s = 0.76$ (95% source-cluster interval $[0.67, 0.84]$). Figure 2 shows the relationship at both resolutions.

This result is the control on which the rest of the argument depends. The machine proof is not arbitrary text set against an organized human object. Both proofs answer to the same type, hypotheses, and library. Provenance appears as a deformation of theorem-specific structure, not as a replacement for it.

4.2 Second, AI provenance tracks an interior shift

At source level, the deformation appears as local decomposition. The median script grows from 23 to 26 lines, a paired change of +2 (95% source-cluster interval $[1, 3]$). The median number of explicit **have** claims grows from 3 to 6. Relative to tactic count, the median paired AI/human ratio is 1.41 (95% source-cluster interval $[1.27, 1.53]$). Equal weighting of the 12 source groups gives a smaller raw **have** contrast of +0.5 $[0, 1.5]$; the pooled direction is therefore not uniform across groups.

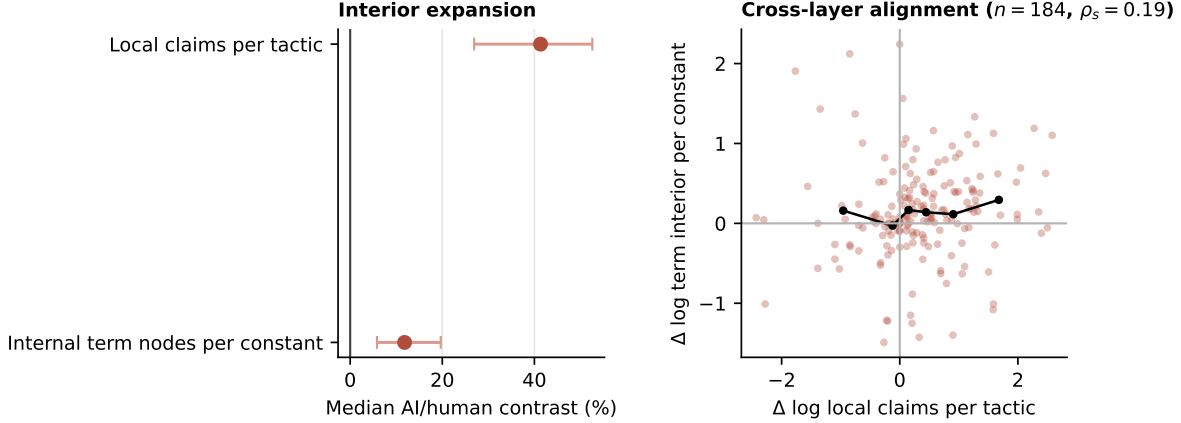


Figure 3: Interior expansion at two resolutions. Left: median paired AI/human contrasts for visible local decomposition and elaborated expression structure. Right: among the 184 pairs observed both ways, the two changes have a modest positive association. Intervals resample source groups; the black line joins six binned means.

Named premise contact shows no corresponding rise with this internal decomposition. The loose and strict source proxies each have paired median difference zero. Their zero-width bootstrap intervals reflect discrete counts, many ties, and only 12 clusters; they are not equivalence tests. The defensible claim is comparative: a clear source-level increase in local claims is not accompanied by a clear increase in the breadth of the named interface.

Elaborated terms show the same asymmetry without relying on source tokens. AI terms contain a median 4,561 nodes, compared with 3,816.5 for human terms; the paired difference is +290 ([20, 529]). The number of distinct constants is nearly unchanged: 200.5 for human terms and 199 for AI terms, with paired difference -1 ($[-3, 0]$). AI terms therefore contain 1.12 times as many non-constant nodes per distinct constant (95% source-cluster interval [1.06, 1.20]). Giving every source group equal weight yields a similar ratio of 1.11 [1.02, 1.21].

The two interior measures are not identical, but they are not unrelated. Among the 184 overlapping pairs, source-level change in local claims per tactic correlates with term-level change in internal nodes per constant at $\rho_s = 0.19$ (95% source-cluster interval [0.04, 0.29]; figure 3). The association is modest, as one would expect when a tactic can elaborate into many expressions. Its importance is methodological: the interior result does not depend entirely on either source syntax or kernel structure.

Taken together, these measurements support one claim. In this corpus, AI proofs often put more internal scaffolding behind a named interface whose breadth changes much less. The data do not identify the mechanism. Explicit subgoal decomposition, coercions, normalization, repeated tactic templates, and less economical theorem APIs are all compatible with the result. Nor does “more interior” mean “less understanding.” It names an architectural contrast, not a mental state.

4.3 Third, the epistemic endpoint remains stable

If each additional expression were an independent chance to fail, the larger AI interior would reduce theorem confidence. The network model does not produce that result. Human and AI belief profiles nearly overlap at all three error rates (figure 4). Averaged across settings, the mean paired AI-minus-human theorem-belief difference is 0.00012 (95% source-cluster interval $[-0.0019, 0.0020]$);

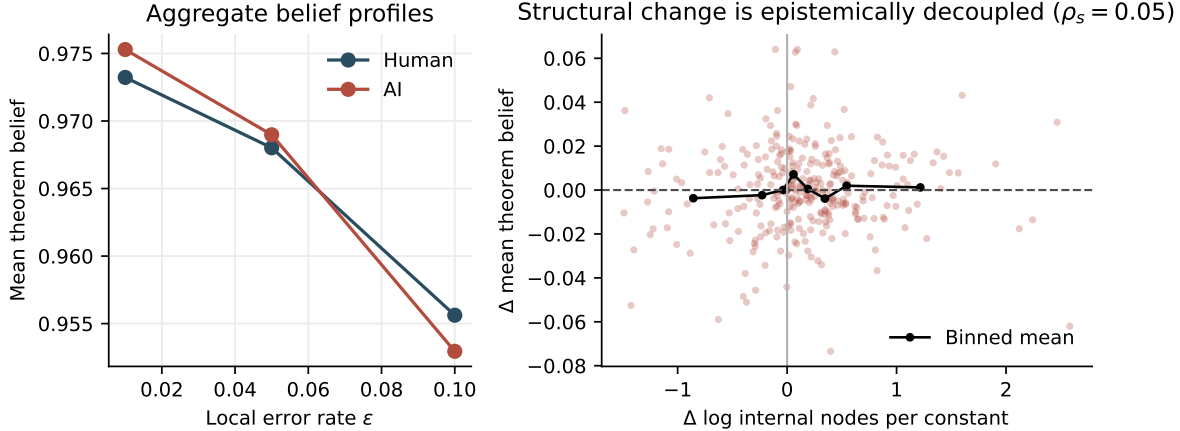


Figure 4: The epistemic endpoint is stable. Left: mean theorem belief for human and AI terms at three local-error rates. Right: pairwise interior expansion is nearly unrelated to pairwise belief change. The black line joins eight binned means and is descriptive, not causal.

the paired median is zero. Every setting-specific cluster interval includes zero.

More importantly, structural change offers little predictive information about epistemic change. Across the 312 pairs, the association between the AI/human log ratio in interior load and the paired change in mean theorem belief is $\rho_s = 0.046$ (95% source-cluster interval $[-0.07, 0.22]$). Within the precision of this sample, proofs with much larger AI interiors show no clear tendency to receive lower modeled theorem belief.

This conjunction is epistemic decoupling. The architecture moves in a consistent direction, but the modeled endpoint does not move with it. The conclusion is stronger than saying that group means happen to be close: we detect no tendency for proof pairs with the largest structural changes to show a correspondingly larger epistemic contrast. It is also narrower than a claim about human belief: the invariance belongs to this network model at this representation.

5 What decoupling means

5.1 The missing scale is the reader’s scale

Experts do not normally inspect every primitive expression with equal attention. They move backward from goals, examine warrants selectively, reconstruct omitted steps, test examples, and treat familiar lemmas as chunks (Weber, 2008; Inglis and Alcock, 2012; Mejía-Ramos and Weber, 2014; Panse et al., 2018). The kernel graph and the reader’s proof are therefore different objects.

Epistemic decoupling makes that difference visible. A certificate can grow internally while a uniform network dynamics assigns its theorem nearly the same confidence. If readers nevertheless find one presentation clearer, that difference must depend on features absent from the model: attention, chunking, names, prior familiarity, explanatory expectations, or navigation across levels. The interface is a plausible place to look because it supplies the handles by which a reader can compress and revisit the interior.

This interpretation extends rather than rejects the phase-transition framework. The original model studies how evidence propagates through a fixed network. A cognitive successor should study how a reader constructs the effective network in the first place. Node granularity, prior belief, and attention should become endogenous to the interaction between proof and reader.

5.2 Confidence cannot select the explanatory proof

A formal certificate establishes derivability conditional on its statement and environment. It does not by itself establish that the formal statement captures the intended mathematics, or that the certificate tracks the reasons a mathematician takes to matter (DeDeo and Duede, *forthcoming*). Proofs can reveal methods, dependencies, and unifying ideas beyond their conclusions (Rav, 1999; Mancosu, 2008).

The paired data add a selection problem. Even after the statement is fixed, multiple checked interiors remain possible. The belief functional studied here supplies no clear criterion for choosing among them. Other virtues must enter: explanatory relevance, modularity at a reader-appropriate scale, stability under library change, and capacity for communal correction. This is consistent with accounts of mathematics in which reliability is secured not only by avoiding error, but by organizing work so that errors can be localized and repaired (Avigad, 2022; De Toffoli, 2021).

5.3 Proof generation becomes proof design

AI proof systems are commonly trained and evaluated first on whether they produce a certificate that passes the kernel (Yang et al., 2023; Ren et al., 2025). That objective has enabled remarkable progress, but it is indifferent among many proofs of the same theorem. The paired comparison shows that the indifference already leaves an architectural trace: AI proofs often enlarge the interior without broadening the named interface.

The engineering response is not indiscriminate compression. A longer proof may expose a useful lemma; a shorter one may hide its main work inside automation. The objective should be proof design: choosing intermediate claims, names, dependencies, and abstraction boundaries so that a valid term becomes a useful mathematical artifact. Proof-refactoring systems already target readability, modularity, maintenance, and reuse (Fu et al., 2026); studies of human–AI formalization likewise point to the value of high-level control (Collins et al., 2026). Epistemic decoupling explains why these goals cannot be reduced to correctness or modeled confidence.

6 Tests that would break the decoupling claim

The present result is model-internal. A decisive next study should hold correctness and theorem fixed while intervening on architecture. For 40–60 Lean theorems, construct four checked presentations: the human-provenance proof, a raw AI proof, a refactored AI proof with designed intermediate lemmas, and a mechanically expanded human proof that preserves the original plan. Randomize the displayed authorship label independently of the underlying proof.

Readers should perform a small set of preregistered tasks: validate the argument, locate a seeded defect in an informal rendering, summarize the main idea, identify essential assumptions, transfer the method to a nearby theorem, and repair the proof after an API change. The primary outcomes are validation accuracy, calibrated confidence, conceptual recall, transfer, and repair cost. Process measures—time, navigation traces, and inspected interface elements—can test the mechanism.

This design can falsify the present interpretation in several ways. Interior expansion may lower human confidence even when the network model is invariant. Refactoring may fail to improve comprehension. Authorship labels may dominate architecture. Or the apparent interface may prove to be the wrong unit once reader behavior is observed. Each failure would locate the scale at which the current model stops being adequate.

A parallel library study should follow accepted human-, AI-, and mixed-provenance contributions through review and later use. Outcomes include citation by later declarations, survival through API

changes, repair time, reviewer effort, and reuse of introduced lemmas. This would test the social half of the thesis: whether an intelligently designed interface helps a certificate become communal mathematical infrastructure (Pantsar, 2026).

7 Limitations

The pairing removes theorem identity as a confound, but it does not identify a pure causal effect of AI authorship. Provenance labels may conceal editing, retrieval, shared data, and selection. The corpus is dominated by competition mathematics, and its 12 source groups are coarse.

The two interior measures are proposed constructs. Source identifiers are lexical heuristics. Term constants are more authoritative, but the graph boundary still depends on Lean, Mathlib, and the elaborator version. Pairwise removal of error-recovery terms excludes 36.8% of eligible pairs and may select on proof complexity. The cross-layer association is positive but modest.

The belief result is not behavioral evidence. Thirty-two chains reduce Monte Carlo noise but do not validate the prior, symmetric error rates, node granularity, or update rule as a model of human reading. The near-zero contrast should be read diagnostically: within this representation and model, a larger AI interior is not accompanied by a detectable epistemic deficit.

Finally, decoupling is not equivalence. Several intervals include zero because the data do not resolve a contrast; they do not prove identity. Conversely, a statistically detectable structural effect need not be cognitively important. The reader experiments above are required to establish that bridge.

8 Conclusion

Holding the theorem fixed changes what can be asked of a proof. Human and AI constructions share a strong theorem-imposed frame, but they distribute work differently within it. AI-provenance proofs in this corpus place more visible and elaborated structure behind each named interface element. The network dynamics that motivated epistemic phase transitions nevertheless assign the two populations nearly the same theorem belief, and the size of the interior change has little association with the belief contrast.

That is epistemic decoupling: architecture changes while a specified epistemic endpoint remains stable. Its importance lies as much in what it withholds as in what it establishes. Correctness and modeled confidence do not select the proof that a reader can understand, a maintainer can repair, or a library can reuse. In a world of abundant certificates, the scientific object is the coupled system of theorem, proof, interface, reader, and community.

A From the original Coq networks to human Lean proofs

This appendix asks a direct question: do the human Lean networks confirm the principal structural and epistemic findings reported by Viteri and DeDeo (2022)? The answer is yes for the conceptual core, but not for every distributional claim.

A.1 Making the comparison commensurable

The original study extracted abstract-syntax trees from Coq proofs, identified repeated claims, and converted the trees into DAGs. The updated human Lean sample applies the analogous pipeline to 33 classic and frontier theorems in Mathlib: proof terms are expanded level by level, structurally

identical expressions are shared, and expansion stops after crossing roughly 10,000 nodes. These expanded term graphs are the appropriate Lean comparison. Declaration graphs, which connect named theorems across a library, have a different boundary and are reported only as a representation sensitivity check.

The historical count requires care. The published article describes 48 Coq proofs. Its archived table supplies 47 matchable exponent entries, while the recovered directories yield 49 analyzable graphs under the updated pipeline. The archive also contains a status list, but it includes aliases and does not specify a unique 48-graph analysis set. We therefore report the actual sample size for each check rather than forcing the archive into one total.

A.2 The fixed tail at $x_{\min} = 10$

The main exponent discrepancy arose from a cutoff choice. The surviving publication script fixed $x_{\min} = 10$; a later audit initially allowed software to choose $x_{\min}$ by minimizing a Kolmogorov–Smirnov distance. Those procedures estimate different parts of the degree distribution. When the fixed-10 convention is restored, 40 of the 44 estimable archived exponents agree with the published values within 0.01, and 43 agree within 0.02; the median absolute discrepancy is 0.003. This is direct provenance rather than a reconstruction from the table alone: the archived fitter hard-codes `cutf=10`, and saved plotting data record that cutoff together with the published Four Color and Gödel exponents.

The convention was not completely uniform. The four smallest published networks do not reproduce under the surviving fixed-10 call. Applying the same discrete fitter at $x_{\min} = 5$ recovers both their exponents and standard errors after rounding. No surviving script contains this fixed-5 branch, so the evidence supports a small-network exception in an earlier or uncommitted version, but does not recover its rule. One further row, Triangle Inequality, differs by 0.012 at $x_{\min} = 10$. These historical exceptions do not enter the cross-assistant comparison below, which applies one declared cutoff uniformly to every estimable Coq and Lean network.

For the tail that begins at reuse degree 10, recovered Coq networks have median $\alpha = 2.143$ (network-bootstrap interval [2.106, 2.199]; $n = 45$). The 33 human Lean term networks have median $\alpha = 2.093$ ([2.041, 2.157]). The median difference is about -0.05 , and a two-sided Mann–Whitney comparison gives $p = 0.124$. Thus a fixed-tail index close to two recurs across the two samples. This is a qualitative replication, not an equivalence test; it answers the explicitly restricted question of what happens above reuse degree 10.

Fixing the cutoff does not, by itself, establish a unique distribution family. We therefore refit both alternatives on exactly the tail $k \geq 10$. Four recovered Coq networks have fewer than ten observations in that tail and are not estimable. Against an exponential, the power law is favored in 43 of the remaining 45 Coq networks and is inconclusive in two; it is favored in all 33 human Lean networks. Against a lognormal, the power law is favored in neither collection. The lognormal is favored in two Coq networks and the remaining 43 are inconclusive; all 33 Lean comparisons are inconclusive. Thus reuse above degree 10 is heavy relative to an exponential, but these data usually cannot distinguish a power law from a lognormal. We do not confirm a universal, cutoff-free pure power law.

A.3 Does local arity follow a Poisson distribution?

The original article’s in-degree claim is straightforward to test once edge orientation and zero degree are made explicit. With edges directed from premise to dependent, in-degree is the number of distinct immediate premises of an expression—its local arity. The archived `histogram_in.rb`

removed degree-zero leaves, so we likewise take positive in-degrees as the primary sample. This choice asks whether local arity is zero-truncated Poisson; it does not ask whether the proportion of leaves is Poisson-generated.

The archived figure was only a visual overlay. Its plotting code set the Poisson parameter equal to the mean positive degree and renormalized over $k \geq 1$. That is not the maximum-likelihood parameter after zero truncation: the fitted λ instead solves

$$\bar{k} = \frac{\lambda}{1 - e^{-\lambda}}.$$

For Four Color, for example, the plotted value was $\bar{k} = 3.154$, whereas the correctly fitted parameter is $\lambda = 2.997$. We fit this model separately to every network and test absolute fit with a discrete Kolmogorov–Smirnov statistic and 199 parametric bootstrap samples, refitting λ in each sample. Rejections below use Benjamini–Hochberg correction within each corpus.

Likelihood comparison uses three natural positive-count alternatives: a geometric distribution, a zero-truncated negative binomial for overdispersion, and a zero-truncated Conway–Maxwell–Poisson distribution for underdispersion (Shmueli et al., 2005). The last contains Poisson as the special case $\nu = 1$ and can concentrate probability on the small set of arities imposed by a term grammar. We compare Akaike information criterion (AIC), charging both flexible models for their additional parameter.

Table 2: Direct tests of the Poisson local-arity claim. Distributions are fitted separately to each network after conditioning on positive in-degree, matching the historical figure. The dispersion ratio is observed variance divided by the variance of the fitted zero-truncated Poisson; one is the Poisson expectation. Best AIC counts network-level winners among Poisson, geometric, negative binomial (NB), and Conway–Maxwell–Poisson (CMP).

Corpus	Networks	Median $\bar{k}$	Dispersion ratio	Poisson rejected	Best AIC
Recovered Coq expanded terms	49	3.498	0.710	49/49	CMP 35; NB 14
Human Lean expanded terms	33	1.968	0.030	33/33	CMP 33/33
Matched human Lean <code>term0</code>	312	2.000	0.0017	312/312	CMP 312/312
Matched AI Lean <code>term0</code>	312	2.000	0.0021	312/312	CMP 312/312

The answer is unambiguous. Poisson is rejected in every Coq, human Lean, matched-human, and matched-AI network. The featured Four Color and Gödel networks are both underdispersed relative to the correctly fitted Poisson (dispersion ratios 0.502 and 0.545) and reject at the bootstrap’s minimum attainable $p = .005$. Thirty-five Coq networks prefer the underdispersed CMP alternative; the other 14 are overdispersed and prefer a negative binomial. All 657 Lean networks prefer CMP. Thus the result is not merely that a different heavy-tailed family wins. Poisson fails in both directions across Coq, and Lean arity is vastly more concentrated than Poisson.

We next fit the suggested special-zero model,

$$P(K = 0) = \pi + (1 - \pi)e^{-\lambda}, \quad P(K = k) = (1 - \pi)e^{-\lambda} \frac{\lambda^k}{k!}, \quad k \geq 1,$$

as well as a hurdle model in which $P(K = 0)$ is completely free. A standard zero-inflated Poisson (ZIP) can accommodate excess leaves; a hurdle Poisson can accommodate either excess or deficient

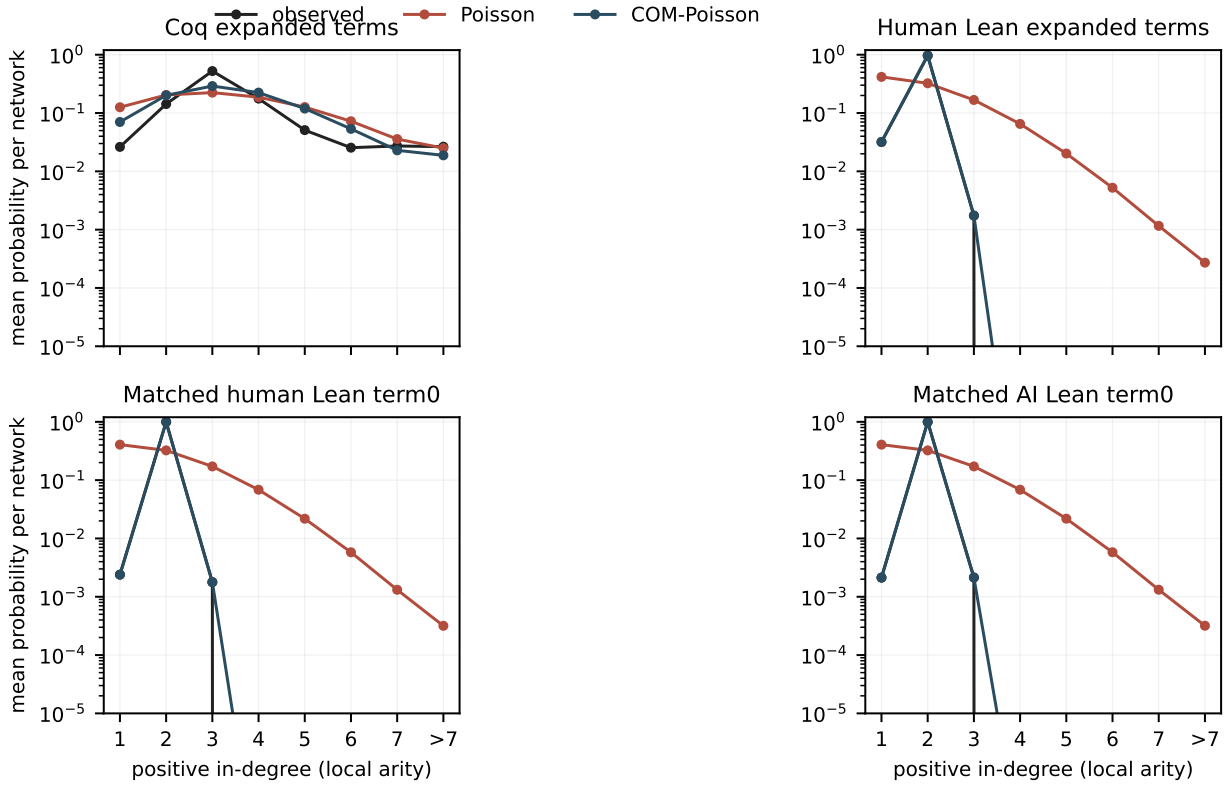


Figure 5: Observed and fitted positive in-degree distributions. Curves average probabilities across networks, giving each proof equal weight after fitting it separately; the final category combines degrees above seven. A Poisson allocates substantial probability to arities that the Lean term grammar almost never produces. Coq is less rigid, but neither Poisson nor a single flexible count family captures all of its constructor mixture.

leaves. Crucially, neither changes the conditional shape for $K > 0$, which remains the zero-truncated Poisson tested above. We nevertheless repeat the refitted parametric-bootstrap test on all nodes and compare full-count AIC against hurdle geometric, NB, and CMP alternatives.

Table 3: Sensitivity to a special zero term. “Boundary” counts fits for which standard ZIP is constrained to $\pi = 0$ because the network has fewer zeros than its Poisson component predicts.

Corpus	Networks	ZIP rejected	Boundary $\pi = 0$	Median π	Best full-count AIC
Recovered Coq expanded terms	49	49/49	1/49	0.171	hurdle CMP 35; hurdle NB 14
Human Lean expanded terms	33	33/33	33/33	0	hurdle CMP 33
Matched human Lean term0	312	312/312	298/312	0	hurdle CMP 312
Matched AI Lean term0	312	312/312	296/312	0	hurdle CMP 312

Giving zero a special term does not change the verdict. Coq has enough excess leaves for an interior ZIP mixture in 48 of 49 networks (median $\pi = 0.171$), but ZIP is rejected in all 49 and is never the best full-distribution model. Lean generally has the opposite problem. Its observed zero fraction is smaller than the Poisson implied by its positive degrees, so the ZIP inflation weight is forced to its zero boundary in 33/33 expanded human networks, 298/312 matched-human networks, and 296/312 matched-AI networks. Hurdle CMP wins all 657 Lean comparisons, and ZIP is rejected in every one. A free leaf probability therefore improves the bookkeeping but cannot repair the Poisson shape among non-leaves.

The same-theorem comparison shows almost no authorship signal. The median share of positive-arity nodes with degree exactly two is 0.9979 for human proofs and 0.9975 for AI proofs; the paired difference is not detectable ($p = 0.185$). The paired Poisson KS difference is exactly zero at the median ($p = .965$). A tiny difference in mean arity is statistically detectable because there are 312 large graphs, but it is only 0.00019 premises per positive-arity node.

The corrected conclusion is therefore structural rather than distributional. Local dependence is small and concentrated, while reuse is broad and heavy-tailed; that contrast survives. But local arity is not Poisson-distributed. In Lean especially, it is largely a signature of the binary application structure and the extractor’s atomic scale. The near identity of human and AI `term0` arity says that both certificates pass through the same elaborator and term grammar, not that their authors use the same cognitive process. Treating leaves as a separately fitted hurdle component does not rescue Poisson, because its positive component is rejected in every network.

A.4 What else confirms

Modularity is especially stable at the term level: median Q is 0.724 in the recovered Coq graphs and 0.717 in the human Lean graphs. At $\epsilon = .01$, both collections reach median theorem belief 1.0; the median belief averaged over all nodes is 0.9939 and 0.9935, respectively. Every recovered Coq and human Lean term graph also has a positive firewall contrast under the updated implementation. The sign agrees with the original claim that modules can localize reversals, but its magnitude should not be compared to the published $\Delta\mathcal{L}_1$: the normalization changed.

Table 4: A claim-by-claim comparison. Medians use the recovered Coq graphs and the 33 expanded human Lean term graphs. “Model” means that the result is a consequence of the specified belief dynamics, not a behavioral measurement.

Claim	Recovered Coq	Human Lean terms	Judgment
Reuse tail above degree 10	$\alpha = 2.143$	$\alpha = 2.093$	Fixed-cutoff confirmation
Heavy tail is specifically a pure power law	Lognormal favored in 2/45; unresolved in 43/45	Lognormal unresolved in 33/33	Not confirmed
High modularity	median $Q = 0.724$	median $Q = 0.717$	Confirmed descriptively
High belief at $\epsilon = .01$	median theorem belief 1.0; mean-node belief 0.9939	median theorem belief 1.0; mean-node belief 0.9935	Confirmed within the model
Positive modular firewalls	positive in 49/49	positive in 33/33	Qualitative confirmation
Poisson-like in-degree	Rejected in 49/49	Rejected in 33/33	Not confirmed; local arity is grammar-constrained

The direct tests in [section A.3](#) replace the earlier tail-family proxy. The broad contrast survives; the specific Poisson claim does not.

The representation check explains why these confirmations must remain qualified. For the same 33 Lean theorems, declaration graphs have median $Q = 0.465$ and fixed-tail median $\alpha = 1.751$, far from the term-graph values. A proof network has no representation-free exponent or modularity. The cross-language agreement is meaningful because the Coq and Lean term pipelines are analogous, not because either reveals a privileged atomic scale.

The overall verdict is therefore precise. Human Lean terms confirm the original paper’s central network claim: fine-grained proof graphs exhibit concentrated reuse, strong modularity, positive firewall structure, and high theorem belief under bidirectional dynamics. They do not justify the stronger folklore that all proof networks instantiate one pure power law with a universal exponent. That correction strengthens the present paper. Epistemic decoupling depends on network robustness, not on exact scale freedom.

B Supplementary estimates and reproducibility

Table 5: Principal same-theorem estimates. Difference rows report AI minus human; ratio rows report AI divided by human. Intervals resample the 12 problem-source groups.

Measure	Human median	AI median	Paired estimate	95% interval
Script lines	23	26	+2	[1, 3]
Explicit have claims	3	6	+1	[0, 1]
Loose source premise proxy	5	4	0	[0, 0]
Strict dotted premise proxy	3	2	0	[0, 0]
Local claims per tactic	.30	.50	$1.41\times$	[1.27, 1.53]
Elaborated term nodes	3,816.5	4,561	+290	[20, 529]
Distinct elaborated constants	200.5	199	-1	[-3, 0]
Constant-node share	.0530	.0456	-.0046	[-.0061, -.0028]
Internal nodes per constant	17.79	20.84	$1.12\times$	[1.06, 1.20]
Mean theorem belief, three settings	—	—	+.00012	[-.0019, .0020]

The same-theorem analysis is implemented in the following script, with its machine-readable summary immediately below:

`code/standalone_paper_analysis.py`

`results/standalone_paper/summary.json`

It uses the corrected paired source table, the clean paired `term0` table, and the 32-run belief output. The 20,000-resample analysis was executed locally and as ORCHARD Slurm job 128546; the summaries were byte-identical. The ORCHARD output archive has SHA-256 `b90b31e99003d40ffdc9e8bad9a027d00d6302cc2798d7c4cf9a520468de48c4`.

The Coq–Lean confirmation is implemented in `code/coq_lean_confirmation.py`; its summary is `results/coq_lean_confirmation_xmin10/summary.json`. Model comparisons were run with one niced core on Akdeniz. The script independently refits all 49 recovered Coq and 33 human

Lean edge lists at $x_{\min} = 10$; its Lean exponent values agree with the prior table to floating-point precision.

The local-arity analysis is implemented in `code/poisson_indegree_appendix.py`; its network-level fits, paired table, calibration curves, and machine-readable summary are under `results/poisson_indegree/`. The script uses the same 49 recovered Coq networks, 33 human Lean expanded term networks, and 312 clean theorem-matched `term0` pairs as the main analyses.

The compact derived tables and code are included with this working directory. Before publication, the raw elaborated graphs and upstream corpus should be deposited with immutable manifests, toolchain versions, and license metadata so that extraction can be reproduced end to end.

References

- AI-MO. NuminaMath-LEAN: A large-scale dataset of competition problems formalized in Lean 4. Hugging Face dataset, <https://huggingface.co/datasets/AI-MO/NuminaMath-LEAN>, 2025. Accessed August 2026.
- Jeremy Avigad. Reliability of mathematical inference. *Synthese*, 199:12389–12416, 2022. doi: 10.1007/s11229-020-02524-5.
- Katherine M. Collins, Simon Frieder, Jonas Bayer, Jacob Loader, et al. Characterizing initial human–ai proof formalization workflows, 2026. arXiv:2606.04273.
- Silvia De Toffoli. Groundwork for a fallibilist account of mathematics. *The Philosophical Quarterly*, 71(4):823–844, 2021. doi: 10.1093/pq/pqaa076.
- Simon DeDeo and Eamon Duede. A correspondence problem for mathematical proof. *Philosophy of Science*, forthcoming. arXiv:2603.13680.
- Yiming Fu, Peixuan Liu, Zichen Wang, and Kun Yuan. Proof-refactor: Refactoring generated formal proofs into modular artifacts, 2026. arXiv:2606.03743.
- Thomas Hubert, Rishi Mehta, Laurent Sartran, Miklós Z. Horváth, et al. Olympiad-level formal mathematical reasoning with reinforcement learning. *Nature*, 651:607–613, 2026. doi: 10.1038/s41586-025-09833-y.
- Matthew Inglis and Lara Alcock. Expert and novice approaches to reading mathematical proofs. *Journal for Research in Mathematics Education*, 43(4):358–390, 2012. doi: 10.5951/jresmetheduc.43.4.0358.
- Paolo Mancosu. Mathematical explanation: Why it matters. In *The Philosophy of Mathematical Practice*, pages 134–150. Oxford University Press, 2008. doi: 10.1093/acprof:oso/9780199296453.003.0006.
- Juan Pablo Mejía-Ramos and Keith Weber. Why and how mathematicians read proofs: Further evidence from a survey study. *Educational Studies in Mathematics*, 85(2):161–173, 2014. doi: 10.1007/s10649-013-9514-2.
- Anja Panse, Lara Alcock, and Matthew Inglis. Reading proofs for validation and comprehension: An expert-novice eye-movement study. *International Journal of Research in Undergraduate Mathematics Education*, 4:357–375, 2018. doi: 10.1007/s40753-018-0077-6.

- Markus Pantsar. How to recognize artificial mathematical intelligence in theorem proving. *Topoi*, 45:707–720, 2026. doi: 10.1007/s11245-025-10164-w.
- Yehuda Rav. Why do we prove theorems? *Philosophia Mathematica*, 7(1):5–41, 1999. doi: 10.1093/phimat/7.1.5.
- Z. Z. Ren, Zhihong Shao, Junxiao Song, Huajian Xin, et al. Deepseek-prover-v2: Advancing formal mathematical reasoning via reinforcement learning for subgoal decomposition, 2025. arXiv:2504.21801.
- Galit Shmueli, Thomas P. Minka, Joseph B. Kadane, Sharad Borle, and Peter Boatwright. A useful distribution for fitting discrete data: Revival of the Conway–Maxwell–Poisson distribution. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 54(1):127–142, 2005. doi: 10.1111/j.1467-9876.2005.00474.x.
- Scott Viteri and Simon DeDeo. Epistemic phase transitions in mathematical proofs. *Cognition*, 225: 105120, 2022. doi: 10.1016/j.cognition.2022.105120.
- Keith Weber. How mathematicians determine if an argument is a valid proof. *Journal for Research in Mathematics Education*, 39(4):431–459, 2008.
- Kaiyu Yang, Aidan Swope, Alex Gu, Rahul Chalamala, Peiyang Song, Shixing Yu, Saad Godil, Ryan J. Prenger, and Anima Anandkumar. Leandojo: Theorem proving with retrieval-augmented language models. In *Advances in Neural Information Processing Systems*, volume 36, 2023.